SC2207: Intro to Databases

Lab 1 Assignment

Lab Group SCSC
Team 1

Members:
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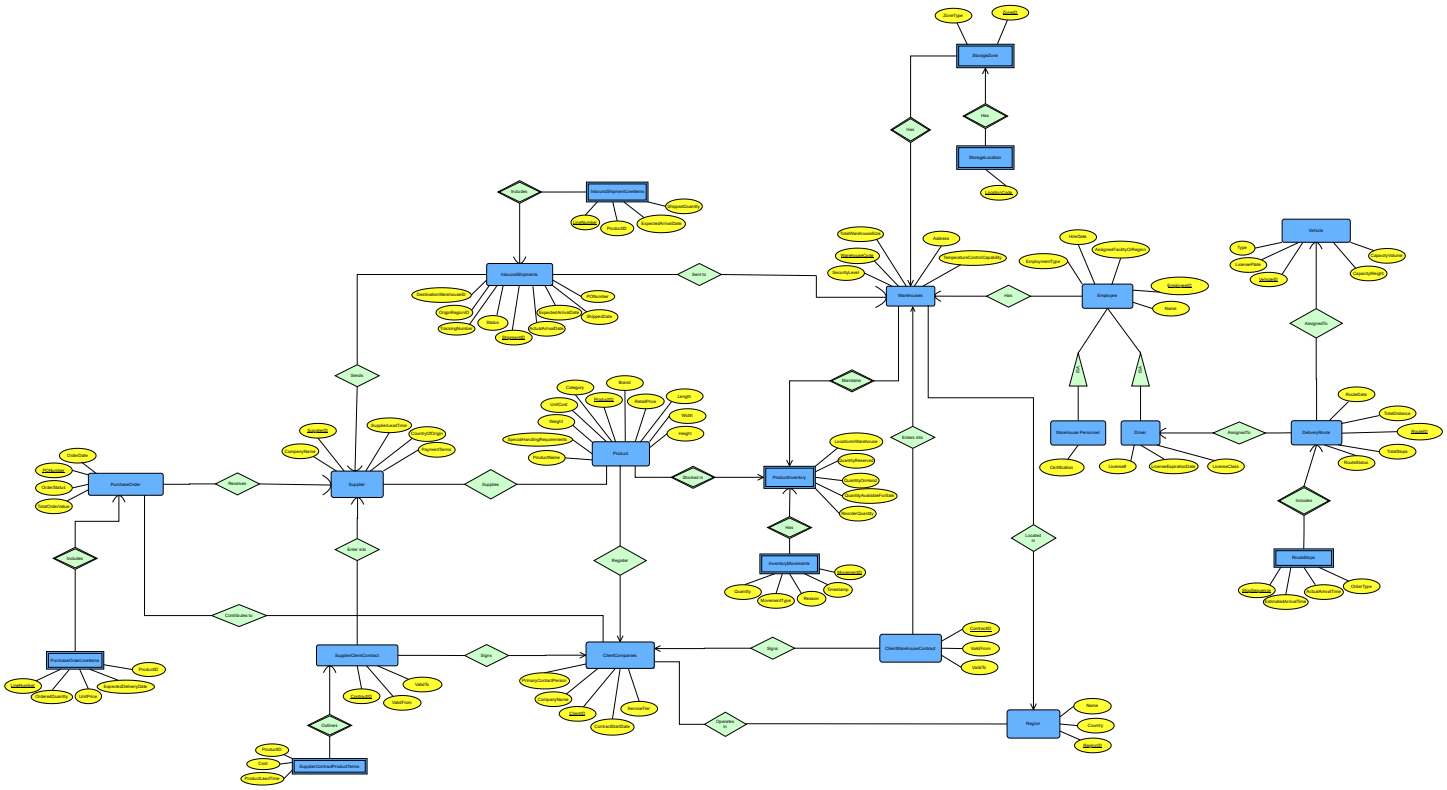
ZoneType
ZoneID
StorageZone
Has
StorageLocation
LocationCode
Has
Includes
InboundShipmentLineItems
ShipmentID
ProductID
ExpectedArrivalDate
ShippedQuantity
InboundShipment
ReceivingWarehouse
ExpectedArrivalDate
ShipmentStatus
ShipperID
CarrierIdentifier
TrackingNumber
ShipmentID
ExpectedDeliveryDate
ActualDeliveryDate
Receives
Warehouses
OperatingHours
SecurityLevel
Address
ManagementContactEmail
Has
Employee
HireDate
AssignedFacilityOfficeID
EmployeeType
EmployeeID
Name
Vehicle
Type
LicensePlate
VehicleID
Capacity
AssignedDriverID
AssignedTo
Warehouse Personnel
Driver
CertificationIDs
License
LicenseExpirationDate
LicenseClass
Assigned To
DeliveryRoute
RouteDate
RouteStatus
RouteID
TotalStops
RouteNotes
Includes
RouteStops
RouteID
StopSequence
RouteActualTime
OrderType
Stores
Stores
Category
Brand
ProductID
RetailPrice
Length
UNSPSC
Width
Weight
Height
SpecialHandlingRequirements
ProductName
Product
Includes In
ProductInventory
ProductWarehouseID
QuantityReserved
QuantityOnHand
QuantityAvailableForOrder
ReorderQuantity
Enters into
Located In
Has
InventoryMovements
MovementID
Quantity
MovementType
Reason
Timestamp
Supplies
Supplies
PurchaseOrder
OrderDate
PurchaseID
OrderStatus
TotalOrderValue
Receives
Supplier
SupplierID
SupplierLeadTime
QualityRating
CompanyName
PaymentTerms
Enter into
Includes
Contributes to
Register
PurchaseOrderLineItems
ProductID
LineNumber
OrderedQuantity
UnitPrice
ExpectedDeliveryDate
SupplierClientContract
Signs
ClientCompanies
Signs
ClientWarehouseContract
ContractID
ValidFrom
ValidTo
SupplierID
ClientID
ValidTo
ValidFrom
PrimaryContactPerson
CompanyName
ClientID
SecondaryEmail
ContractStartDate
Operates in
Regions
Name
Country
RegionID
Outlines
SupplierContractProductTerms
ProductID
Date
ProductLeadTime

Assumptions

1. The **quantity on hand** is **equal** to the **total stock in the warehouse**.
2. The **sum of Reserve and forSale** is **equal** to the **quantity on hand.**
3. Every client can have **only one contract** with one supplier.
4. **Supplier lead time** is the **time taken from when the order is placed and when it is delivered to the warehouse.**
5. **Product lead time** is the **time taken to manufacture a particular product**.

Weak Relationships

Relationship	Entities Involved	Explanation
Outlines	SupplierClientContract - SupplierContractProductTerms	All weak relationships tie a weak entity to its owner. They enforce existence dependency and unique identification. They also keep the diagram normalized and reduce redundant data.
Includes	PurchaseOrder - PurchaseOrderLineItems, InboundShipments - InboundShipmentLineItems, DeliveryRoute - RouteStops	
Has	Warehouses - StorageZone, StorageZone - StorageLocation ProductInventory - InventoryMovements	
StockedIn	Product - ProductInventory	
Maintains	Warehouse - ProductInventory	

Weak Entities

Entity	Explanation
POLineItem, ShipmentLineItem	They both exist only as part of Purchase Order and Shipment, respectively. Both can be identified by their LineNumber attribute.
StorageLocation, StorageZone	StorageLocation cannot exist without StorageZone, which in turn, cannot exist without the Warehouse entity. They can be identified by LocationCode and ZoneID respectively.
SupplierContractProductTerms	SupplierContractProductTerms is a byproduct of SupplierClientContract and therefore depends on SupplierClientContract.
RouteStops	RouteStops cannot exist without an instance of the Delivery Entity.
InventoryMovements	Inventory movements can only take place once the product is in the warehouse and as such it forms a weak entity.

APPENDIX C: INDIVIDUAL CONTRIBUTION FORM

Full Name	Individual Contribution to Lab 1 Submission	Percentage of Contribution	Signature
Madhav Raghu Anantharam	Write up, fact check, resolved inconsistencies	20%	
Samarth Bose	Created diagram skeleton with entities, relations, attributes. Reviewed inconsistencies	20%	
Serena Dai Yi	Reviewed ER diagram and writeup	20%	
Clarice Kevin Stansilas	Adding attributes, diagram presentation	20%	
Singh Ashna	Adding relationships to ER diagram, final checks	20%	

Full Name	Individual Contribution to Lab 3 Submission	Percentage of Contribution	Signature

Full Name	Individual Contribution to Lab 5 Submission	Percentage of Contribution	Signature

APPENDIX D: USE OF AI TOOL(S) IN LAB WORK

Each team member should indicate either A or B:

A. I affirm that my contribution(s) to the lab work is my own, produced without help from any AI tool(s).
B. I affirm that my contribution(s) to the lab work has been produced with the use of AI tool(s).

Team member (full name)	Signature	Date	A or B
Madhav Raghu Anantharam		5/2/25	A
Samarth Bose		5/2/25	A
Serena Dai Yi		5/2/25	A
Clarice Kevin Stansilas		5/2/25	A
Singh Ashna		5/2/25	A

By signing this form, you declare that the above affirmation made is true and that you have read and understood NTU's policy on the use of AI tools.

If any team member answered B, the team member(s) must indicate and replicate the table below for every instance that AI tool(s) is used:

Name of AI tool	< For example, ChatGPT >
Input prompt	< Insert the question that you asked ChatGPT >
Date generated	
Output generated	< Insert the response verbatim from ChatGPT >
Output screenshots	
Impact on submission	< Briefly explain which part of your submitted work was ChatGPT's response applied >